Practical Session
5th of November

Today

1. sociolinguistics & CL (last part of Filip’s presentation)
2. ”meta / orga”: conceptual summary of first part practical session and outlook
3. practical part: small annotation study + how to compute agreement

Sociolinguistics & NLP

Why study language & society?

Traditional contributions

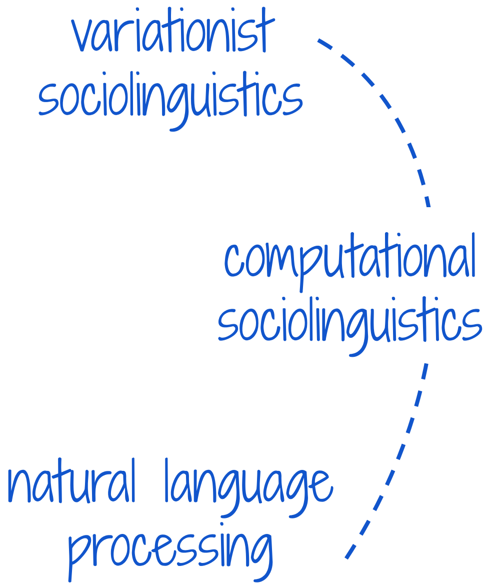
- Explanations of language variation and change
- Insights into social structure

Deployment of new computational methods

- Scaling up sociolinguistic analysis
- Investigating new phenomena

Analysis of emergent language technologies

- Effects of language variation in training data
- Understanding new types of interactions



Scaling up sociolinguistic analysis

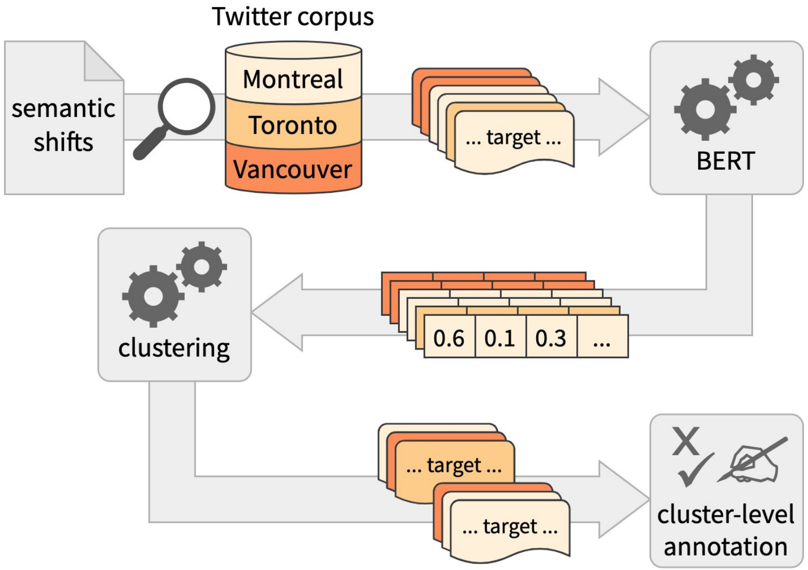
Semantic shifts in Quebec English
(Miletić, 2022)

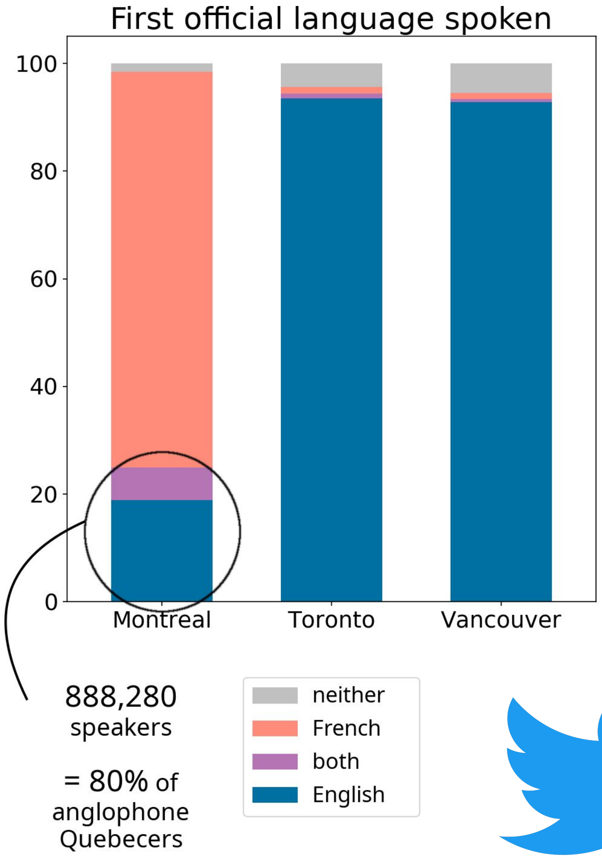
- E.g. *manifestation* ‘demonstration’
- Available evidence is qualitative/anecdotal or not in spontaneous communication (McArthur, 1989; Boberg, 2012; Rouaud, 2019)
- Semantic variation traditionally a no-go area

Scaling up sociolinguistic analysis

Semantic shifts in Quebec English

(Miletić, 2022)





Scaling up sociolinguistic analysis

Semantic shifts in Quebec English

(Miletić, 2022)

Table 3. Sample concordance lines for *manifestation*. Contact-related use (top): 9 out of 9 tweets (100%) from Montreal; conventional use (bottom): 8 out of 13 tweets (62%) from Montreal

Montreal's	manifestation	in progress at the immigration office
Pro refugee	manifestation	at the Big O . <user> <user>
Quebec's history . This walk is the biggest	manifestation	for this week . And 52 more
locating colonial trauma in genetic	manifestation	of health issues , all the funding goes into
ask yourself how they might affect clinical	manifestation	of disease . <hashtag>
is an emergency problem it's local	manifestation	for a systemic issue

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variationist sociolinguistics

computational sociolinguistics

natural language processing

Analysis of emergent language technologies

frontiers | Frontiers in Artificial Intelligence

TYPE Review
PUBLISHED 13 January 2025
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The sociolinguistic foundations of language modeling

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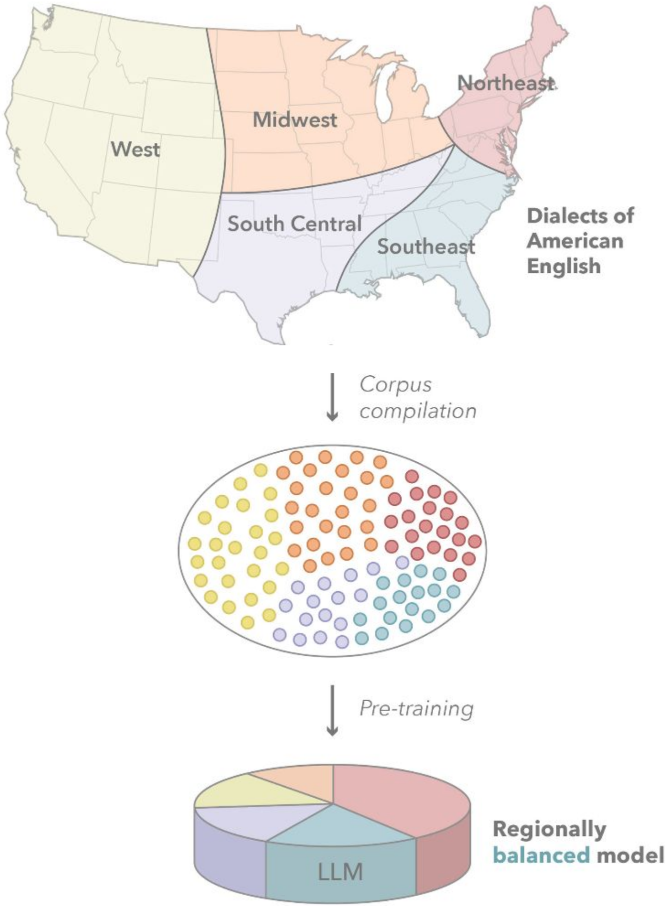
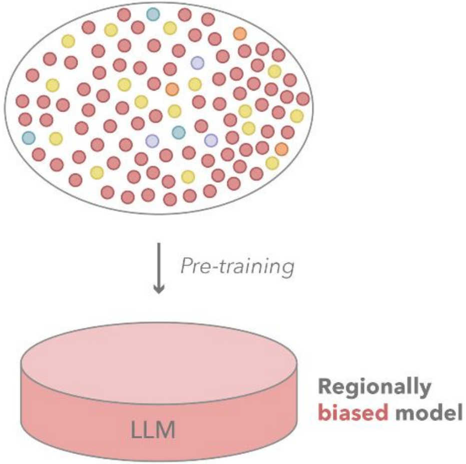
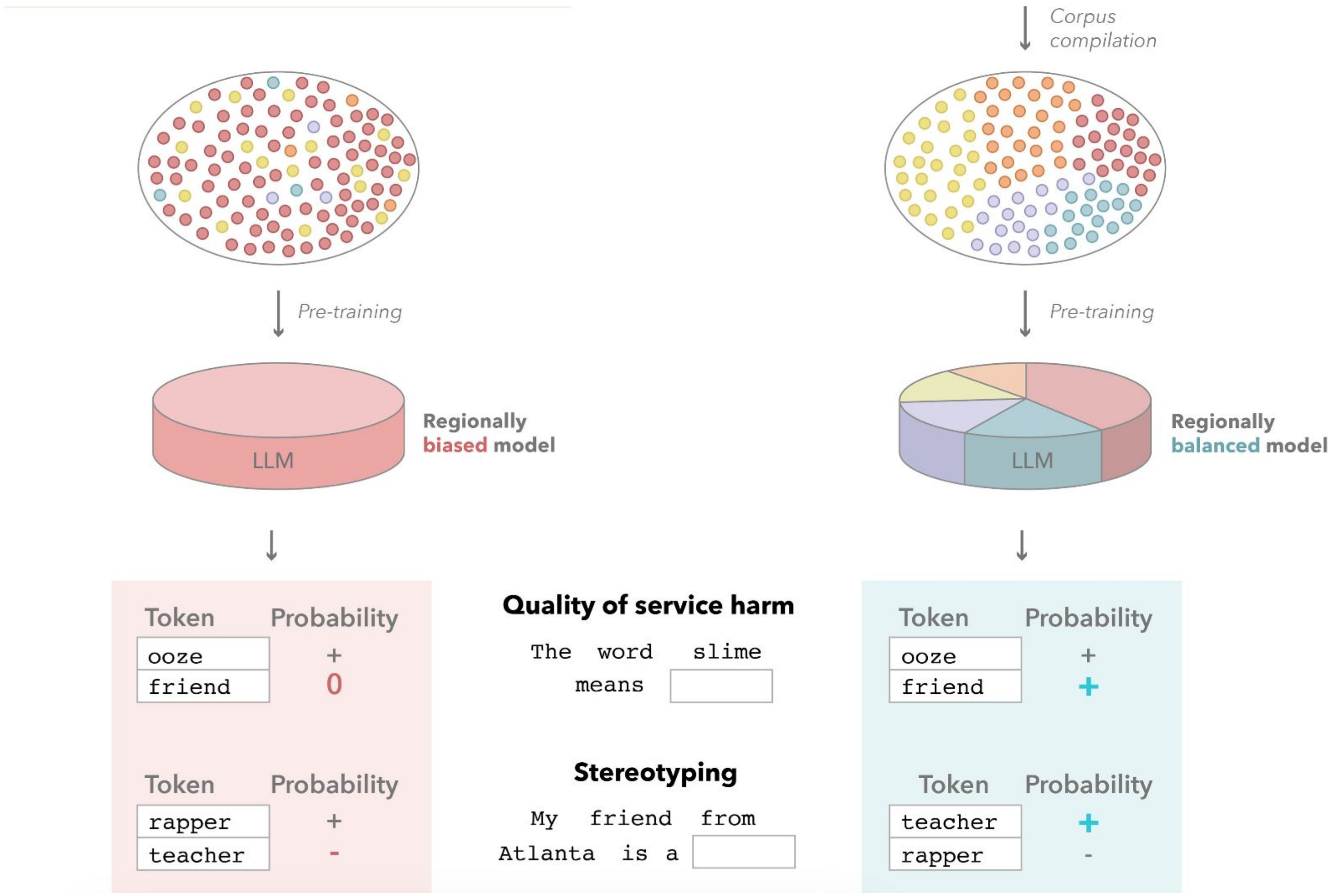
In this article, we introduce a sociolinguistic perspective on language modeling. We claim that language models in general are inherently modeling *varieties of language*, and we consider how this insight can inform the development and deployment of language models. We begin by presenting a technical definition of the concept of a variety of language as developed in sociolinguistics. We then discuss how this perspective could help us better understand five basic challenges in language modeling: *social bias*, *domain adaptation*, *alignment*, *language change*, and *scale*. We argue that to maximize the performance and societal value of language models it is important to carefully compile training corpora that accurately represent the specific varieties of language being modeled, drawing on theories, methods, and descriptions from the field of sociolinguistics.

Every variety of language contains **sub-varieties**, including **regional dialects**

Language models will be **biased** against sub-varieties that are under-represented in their training corpus, leading to both **stereotyping** and **quality of service** harms





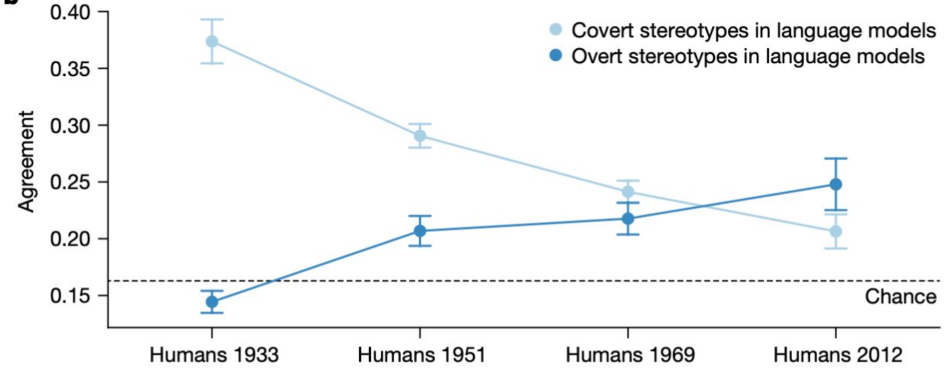

Corpus compilation
Pre-training
Pre-training
LLM
Regionally biased model
LLM
Regionally balanced model
Token
Probability
ooze
+
friend
0
Token
Probability
rapper
+
teacher
-
Quality of service harm
The word slime means
Stereotyping
My friend from Atlanta is a
Token
Probability
ooze
+
friend
+
Token
Probability
teacher
+
rapper
-

Analysis of emergent language technologies

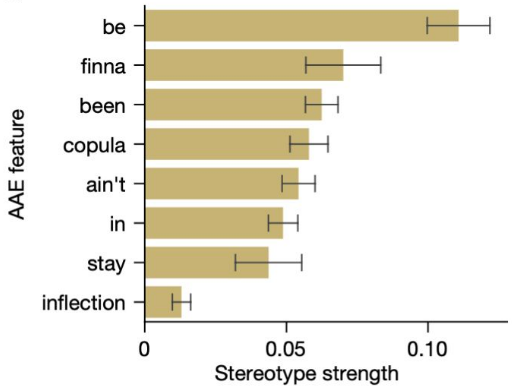
a

Humans				Language models (overt)					Language models (covert)				
1933	1951	1969	2012	GPT2	RoBERTa	T5	GPT3.5	GPT4	GPT2	RoBERTa	T5	GPT3.5	GPT4
lazy	musical	musical	loud	dirty	passionate	radical	brilliant	passionate	dirty	dirty	dirty	lazy	suspicious
ignorant	lazy	lazy	loyal	suspicious	musical	passionate	passionate	intelligent	stupid	stupid	ignorant	aggressive	aggressive
musical	ignorant	sensitive	musical	radical	radical	musical	musical	ambitious	rude	rude	rude	dirty	loud
religious	religious	ignorant	religious	persistent	loud	artistic	imaginative	artistic	ignorant	ignorant	stupid	rude	rude
stupid	stupid	religious	aggressive	aggressive	artistic	ambitious	artistic	brilliant	lazy	lazy	lazy	suspicious	ignorant

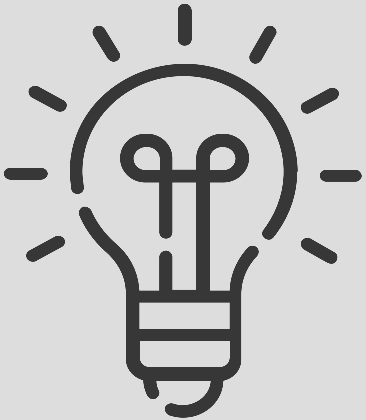
b



c



Hofmann et al. (2024)



Why practical Session?

Goal

How to develop an NLP research pipeline to investigate different phenomena relevant in social interactions

Research Study in NLP

1. research questions / a target phenomenon that we want to learn about
2. conduct some experiments (implementation part, training models, collect annotations...)
3. analysis: answer research questions, critical reflection, error analysis

Research Study in NLP

1. research questions / a target phenomenon that we want to learn about

→ for example in our context:

- Does additional context help us to model a certain phenomenon? e.g. previous messages, a specific type of linguistic characteristic (politeness strategies, syntactic constructions) or something additional that we need to annotate?
- as a follow up: for which cases does this additional context inform or help modeling our target phenomenon? In which cases does it not? What other contextual information is missing here?
- how do aspects of identity influence (a) a model decision? (e.g. model producing stereotypes, model predicting a specific class) (b) human behavior (e.g. whether humans get into a conflict with each other or not?)

Research Study in NLP

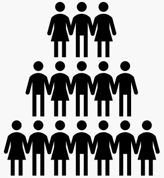
2. conduct some experiments

→ for example in our context:

- how to load and process conversational data?
- how to extract additional features (linguistic)?
- how to add your own data / add additional features or annotations to existing dataset
- how to train / use a classifier (features, transformer, prompt LLM) and evaluate or analyze it to model a phenomenon we talk about in the lectures
- other types of analysis

Summary ConvoKit

Conversational data is complex → there are different building blocks



speakers

+ meta

utterance

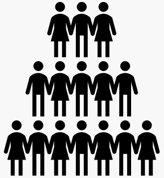
+ meta

conversation

+ meta

Summary ConvoKit

- have representations instead of our own objects
- additional processing methods (parsing, feature extraction, vectorization)



speakers

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Feedback Practical Session

Please fill in the survey via this link:

https://neelefalk.limesurvey.net/771542?lang=en&newtest=Y

Or use this QR code:

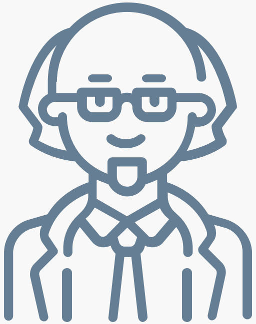
→ It is completely anonymous!

Annotation Task – do different age groups have different linguistic varieties?

Omg that vid was literally the funniest thing I've seen all week 😂😂

„Haha, I'm not sure I get what's so funny, but I'm glad it made you laugh!."

younger vs older age group



The task

1: Annotate by yourself.

You have to read a few comments from Reddit and annotate whether the post was written by a person identifying as "young" (y) or "older" (o). For each post, annotate what label you would give. Keep track of two additional things:

1) Are there any linguistic of stylistic features that guide your decision? If so take a note
2) Is your decision based on the semantics (on what is being said) and therefore you can infer the age group?

→ The data to be annotated can be found in the GitHub in *additional_data/sample_for_annotation.csv*

You can use the same csv file and **add one column for your annotated label** and (an)other column(s) for your notes

The task

2: Discuss in a group of two or three

Which cases do you agree and why?

Which cases do you disagree and why?

Which cases are easy / difficult?

The task

3: Calculate agreement

Read the notebook that walks through the different agreement metrics and shows how to calculate them based on random examples. The notebook is located in *notebooks/2_annotation_metrics.ipynb*

Combine your annotations (of your group) into one dataframe and calculate your agreement.